



Extra comparison with the Decision Tree algorithm

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Yes — **you should include a tree-based algorithm.**

In your setup, it is not just useful; it makes the comparison much stronger.

Why it matters

You are really comparing **three different things**:

- **CarenR** as a rule discovery approach for your problem
- **RIPPER** as an interpretable **rule-based classification** benchmark
- **tree-based models** as a more standard **predictive baseline**

Without a tree model, the reader may ask:

“Are the rule methods good only relative to each other, or are they also competitive against a standard supervised learner?”

A tree-based benchmark answers that.

What each method gives you

Method	Type	Target	Main role in thesis
CarenR	Rule discovery	continuous-friendly / your native setup	Main proposed rule method
RIPPER	Rule-based classifier	discretized target	Interpretable benchmark rule learner

Method	Type	Target	Main role in thesis
Decision Tree / tree-based regressor	Tree model	continuous target	Standard baseline for predictive comparison

This creates a very clean story:

- **CarenR vs RIPPER** → comparison of interpretable rule learners
- **CarenR vs tree model** → comparison of rule discovery vs conventional ML
- **RIPPER vs tree classifier** → optional classification comparison if you want symmetry

My recommendation

You should include at least one tree-based model for the continuous target

Preferably:

- **Regression Tree**
- and optionally a stronger ensemble such as:
 - **Random Forest**
 - **Gradient Boosting**

But if your thesis focus is **interpretability**, then the most important tree baseline is:

Decision Tree Regression

because it is:

- interpretable
- naturally suited to continuous targets
- easy to explain
- a fair comparison against rule-based approaches

Best minimal setup

If you want to keep the thesis focused and not explode the number of experiments, I would do:

Main models

1. **CarenR**
2. **RIPPER**
3. **Regression Tree**

That is already a strong setup.

Stronger setup

If you want one predictive upper-bound style benchmark:

1. **CarenR**
2. **RIPPER**
3. **Regression Tree**
4. **Random Forest Regressor** or boosted trees

This lets you distinguish between:

- **interpretable predictive models** → CarenR / RIPPER / regression tree
 - **high-performance black-box-ish benchmark** → random forest / boosting
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Why a regression tree is especially important in your case

Because your target is **continuous wine production**.

That means:

- RIPPER needs target discretization
- CarenR may preserve more target information
- a **regression tree** works directly on the continuous target

So a regression tree is the cleanest “classical ML” comparator to CarenR.

If CarenR performs reasonably close to the regression tree while remaining highly interpretable, that is a very good result.

If the regression tree clearly outperforms CarenR, you can still argue that:

- CarenR provides domain rules

- those rules may have explanatory value even if predictive power is lower

That is academically solid.

My strongest suggestion on model choice

Include:

- **Regression Tree** as mandatory baseline

Optional:

- **Random Forest / XGBoost** as performance benchmark

Do not overdo:

You do **not** need many tree methods unless your thesis is mainly about predictive modeling.

Your thesis seems more about:

- rule discovery
- interpretability
- comparison of rule structures
- prediction from rules

So one interpretable tree baseline is enough, and maybe one stronger ensemble if you want a performance ceiling.

Suggested experiment logic

Block 1 — Rule discovery comparison

- CarenR
- RIPPER

Block 2 — Predictive comparison

- CarenR-based prediction
- RIPPER-based prediction
- Regression Tree

- optional Random Forest

Block 3 — Rule interpretation

- compare discovered meteorological patterns
- discuss whether RIPPER and CarenR align
- discuss whether tree splits support similar structure

That is a very coherent thesis design.

Thesis-wise answer

Yes, because a tree-based model gives you:

- a **standard ML baseline**
 - a method that handles the **continuous target directly**
 - an additional interpretable structure for comparison
 - stronger evidence that your conclusions are not dependent only on rule learners
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Final recommendation

If you include only one extra method, make it:

Regression Tree

If you include two, make them:

Regression Tree + Random Forest

That would be my strongest current recommendation.

I can also help you build the **final experiment matrix** with:

- method
 - target type
 - input representation
 - evaluation metrics
 - expected role in thesis.
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Yes, you can compare the rules extracted from a decision tree with the rules from RIPPER and CarenR —

but model accuracy alone does not guarantee that the rules are comparable

Let me explain why.

1. Yes — trees also produce rules

A decision tree can always be converted to rules.

Example tree:

```
IF rainfall < 20
  IF temperature > 30
    → high production
  ELSE
    → medium production
ELSE
  → low production
```

becomes rules:

```
Rule 1:
rainfall < 20 AND temperature > 30 → high production

Rule 2:
rainfall < 20 AND temperature ≤ 30 → medium production

Rule 3:
rainfall ≥ 20 → low production
```

So technically:

- **tree paths = rules**

That makes comparison possible.

2. But tree rules are not learned the same way

This is the key difference.

Tree learning

Uses **divide-and-conquer**

It splits the space hierarchically.

Rules correspond to **paths from root to leaf**.

Each rule is **dependent on previous splits**.

RIPPER / CarenR

Use **separate-and-conquer**

Rules are learned **independently**.

Each rule directly targets a subgroup.

3. Why accuracy alone is not enough

A tree can have **very good predictive accuracy**, but the rules may not correspond to meaningful patterns.

Example:

Tree rule:

```
temp > 28.3  
AND rainfall ≤ 17.4  
AND humidity > 63.8  
AND wind ≤ 2.7
```

This may predict well but:

- may be very specific
- may depend on earlier splits
- may not generalize well as a rule

So predictive accuracy $\neq$ rule quality.

4. What should you compare instead?

For your thesis, the comparison should include **three dimensions**.

Predictive performance

Example metrics:

- RMSE
- MAE
- classification accuracy (for RIPPER)

This tells you how well the models predict.

Rule structure similarity

Check whether algorithms discover **similar meteorological patterns**.

Example:

CarenR rule:

```
high temperature + low rainfall → high production
```

RIPPER rule:

```
temp > 28 AND rainfall < 20 → high production
```

Tree split:

```
temp > 27  
rainfall < 21
```

If these align, it is strong evidence.

Rule interpretability

Compare:

- number of rules
- rule length

- coverage
- stability

Example table:

Model	# Rules	Avg rule length	Coverage
CarenR	6	3	78%
RIPPER	8	4	72%
Tree	12	5	100%

5. Important limitation of tree rules

Tree rules are often **not minimal**.

Example:

Tree rule:

```
A AND B AND C → class
```

But maybe:

```
A AND C → class
```

is sufficient.

Tree rules inherit unnecessary conditions because of the tree structure.

Rule learners usually produce **shorter rules**.

6. So what does a good tree fit tell you?

If the tree has good predictive performance, it tells you:

- the feature set contains predictive signal
- relationships exist in the data
- rule discovery methods should also find meaningful patterns

But it does **not guarantee that tree rules are the best rules**.

7. What you should say in your thesis

A good phrasing is:

Decision trees were included as a benchmark predictive model. Although tree structures can be translated into rule sets by converting root-to-leaf paths into IF–THEN rules, the learning strategy differs from rule induction algorithms such as RIPPER and CarenR. Tree rules emerge from hierarchical partitioning of the feature space, whereas rule learners directly search for predictive subgroups. Therefore, tree-derived rules provide a useful reference for structural comparison, but they are not necessarily optimized for minimal or independent rule discovery.

That is academically correct.

8. Final interpretation

You should use the tree model to answer:

- 1 Is there predictive signal in the data?
 - 2 Do the tree splits support similar relationships found by the rules?
 - 3 Are rule learners able to extract simpler explanations of the same phenomena?
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9. Your reasoning corrected slightly

Your reasoning was:

If the fit accuracy is good, then the rules should be comparable.

Corrected version:

If the predictive accuracy is good, it indicates that meaningful relationships exist in the data. These relationships may appear both in tree splits and in rules discovered by rule learning algorithms, allowing structural comparison of the patterns identified by different methods.

If you want, I can also show you **one very powerful evaluation table format** that many rule-learning papers use to compare **RIPPER, trees, and subgroup**

discovery algorithms. It will fit perfectly in your thesis results section.